



M-Tech Production & Marketing

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Week 2- **Experiment 6 Report**

Internship Program — Batch 2026

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1. Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program** on **09 July 2026**. During today's session, I successfully studied and completed **Experiment 06: Unsupervised Learning – Clustering and Principal Component Analysis (PCA)** from the **Artificial Intelligence & Machine Learning Lab Manual**. The primary objective of this experiment was to understand how unsupervised Machine Learning algorithms identify hidden patterns within unlabeled datasets. I also gained practical experience in applying the **K-Means Clustering** algorithm, determining the optimal number of clusters using the **Elbow Method**, and performing **Principal Component Analysis (PCA)** for dimensionality reduction using the **Scikit-learn** library.

2. Objectives

The primary objectives of today's learning session were:

- To understand the fundamental concepts of **Unsupervised Machine Learning**.
- To learn how the **K-Means Clustering** algorithm groups similar data points into clusters.
- To determine the optimal number of clusters using the **Elbow Method**.
- To understand the concept and importance of **Principal Component Analysis (PCA)**.
- To reduce high-dimensional datasets into fewer dimensions while preserving important information.
- To visualize clustered and transformed datasets for better interpretation and analysis.

3. Task Description

Today's task involved studying **Experiment 06** from the AI/ML Lab Manual, which focused on **Unsupervised Learning** techniques. Unlike supervised learning, unsupervised learning works with datasets that do not contain predefined labels, allowing algorithms to discover hidden structures and relationships within the data.

The experiment introduced the **K-Means Clustering** algorithm, which groups similar data points into clusters based on their characteristics. I also learned how the **Elbow Method** helps determine the most appropriate number of clusters by analyzing the relationship between the number of clusters and the corresponding inertia values.

In addition, I studied **Principal Component Analysis (PCA)**, a dimensionality reduction technique that transforms high-dimensional datasets into a smaller number of principal components while preserving most of

the important information. This process simplifies data analysis, improves visualization, and enhances computational efficiency.

4. 4. Work Done

During today's internship session, I successfully completed **Experiment 06** from the AI/ML Lab Manual.

I began by studying the principles of **Unsupervised Machine Learning** and learned how clustering algorithms identify hidden patterns within unlabeled datasets. I explored the working mechanism of the **K-Means Clustering** algorithm and applied it to sample datasets to observe how similar data points are automatically grouped into different clusters based on their characteristics.

Next, I practiced using the **Elbow Method** to determine the optimal number of clusters. By analyzing inertia values across different numbers of clusters, I learned how to identify the point where increasing the number of clusters no longer provides significant improvement in model performance.

I then studied **Principal Component Analysis (PCA)** and understood how it reduces the dimensionality of datasets while preserving the maximum possible information. Using the **Scikit-learn** library, I transformed a multi-dimensional dataset into two principal components, making it easier to visualize and interpret complex data.

Finally, I explored the practical applications of clustering and dimensionality reduction in real-world Machine Learning problems such as **customer segmentation, exploratory data analysis, pattern recognition, and data visualization**. These practical exercises significantly improved my understanding of unsupervised learning techniques.

5. 5. Challenges Faced

The concepts covered in **Experiment 06** were informative and practical; however, a few topics required additional attention. Initially, understanding how clustering algorithms identify patterns without labeled data was slightly challenging. Interpreting the **Elbow Method** graph to determine the optimal number of clusters also required careful observation and analysis.

Furthermore, understanding how **Principal Component Analysis (PCA)** reduces the dimensionality of datasets while preserving the most significant information required additional practice. By reviewing theoretical concepts, experimenting with sample datasets, and analyzing graphical visualizations, I developed a much clearer understanding of these techniques.

6. 6. Key Learnings

During today's learning session, I achieved the following outcomes:

- Developed a strong understanding of **Unsupervised Machine Learning** concepts.
- Learned how the **K-Means Clustering** algorithm groups similar data points into meaningful clusters.
- Understood the purpose and implementation of the **Elbow Method** for selecting the optimal number of clusters.
- Gained practical experience in applying **Principal Component Analysis (PCA)** for dimensionality reduction.
- Learned how dimensionality reduction improves data visualization, computational efficiency, and Machine Learning performance.
- Improved my ability to analyze, visualize, and interpret unlabeled datasets using modern Machine Learning techniques.

7. 7. Conclusion

The successful completion of **Experiment 06** provided valuable theoretical knowledge and practical experience in **Unsupervised Machine Learning**. Through the implementation of **K-Means Clustering** and **Principal Component Analysis (PCA)**, I gained a deeper understanding of data grouping, dimensionality reduction, and data visualization techniques.

The knowledge and practical skills acquired during this experiment have significantly enhanced my confidence in working with unlabeled datasets and applying unsupervised learning methods to real-world Artificial Intelligence and Machine Learning problems. This learning will serve as a strong foundation for future experiments involving advanced clustering techniques, feature engineering, and more sophisticated AI applications throughout the internship program.

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10. Intern Information

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